

Pokémon Team Builder — Take-Home Case Study

Build an end-to-end Pokémon team-building application with a frontend UI, a backend API, tests, and storage. You may use any technologies, including AI.

About Pokémon

Pokémon is a franchise built around collectible creatures, each with a name, one or two elemental types (Fire, Water, Grass, Electric, etc. - 18 types), and base stats like HP and attack. Players build teams of up to 6 Pokémon and battle them against other teams.

Types have strengths and weaknesses against each other (e.g., Water beats Fire, Fire beats Grass). A well-built team usually has a mix of types, so it isn't easily countered by any single opposing type. A chart of type effectiveness can be found here: <https://pokemondb.net/type>.

PokéAPI Reference

Base URL: <https://pokeapi.co/api/v2/> — full documentation at <https://pokeapi.co>. Key endpoints:

- List all Pokémon: <https://pokeapi.co/api/v2/pokemon>
 - Individual Pokémon detail: <https://pokeapi.co/api/v2/pokemon/{id}/>
 - Name: name
 - Types: types[] → type → name
 - Stats: stats[]
 - Moves: moves[]
 - Sprite: sprites → front_default
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Task 1: Team Builder

1. Display all Pokémon in a responsive grid showing name, type(s), stats, and sprite image.
 2. Allow users to build a team of up to 6 Pokémon. Order matters; users can add, remove, and **reorder**. Persist all team information so a user can access their teams **across sessions**.
 3. Allow users to create, delete, and update multiple teams using the team builder UI.
 4. Given a Pokémon team, create an optimal counter team of **equal size**. Open-ended. Show your creativity here!
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Task 2: Data Change Notifications

PokéAPI data can change on rare occasions.

1. Create a backend job to scan for changes in any data used by the team builder.
2. If a user's Pokémon were involved in a change within the last week, display a frontend alert to that user describing what changed.
3. Prove that this feature works in a demo.

Task 3: Technical Document

1. Create a one-page README.md on key design decisions and assumptions, showing your technical acumen and communication skills.

Deliverables

- All code and readme in a shareable repository.
- Frontend hosted at a public URL.